

BAZIL UDDIN KHAN

24K-0559

BSCS-1H

REAL WORLD APPLICATION OF NUMBER SYSTEMS :

1). BINARY :

1. Its of base 2.
2. Its represented with 2 only 2 digits 0 and 1.
3. Through binary digits or numbers we can perform different bitwise operators like and , or, not,xor etc.
- 4 . In computer memory or hardware binary numbers are used.
5. It's a low level language.

2). OCTAL :

1. Its of base 8.
2. Digits from 0 to 7.
3. Different operating system requires file permission in octal form.

3). HEXADECIMAL :

- 1). Its base 16.
- 2). Consists of 4 bits. Like 1001.
- 3). Different letter like 10 represents A and till f which represents 15.

HOW BINARY IS USED IN MACHINE-LEVEL PROGRAMMING :

In machine language binary is used directly to represent instructions and data. Every instruction is executed by the CPU is been encoded in binary form. For instance, the instruction to add two numbers might be represented in binary as follows or converting denary numbers into binary numbers etc.

Instruction:

00100111

This could translate to an operation code (opcode) that instructs the CPU to add the values stored at specified memory addresses. So it could translate it and get values. The answer will be 39.

LIKE addition of 5 and 2 in binary will be:

101 + 010 = 101

Advantages and Disadvantages of binary :

Advantages:

- It helps to easily identify which number is open represented through at 0/1.
- Easily represented

Disadvantages:

- Long sequence of binary digits could place problems.

Hexadecimal Number System Roles in Memory Addressing :

Hexadecimal is particularly useful in memory addressing as it allows for more manageable and reasonable representation of large binary digits. As Each hexadecimal digit corresponds to four binary (bits), that simplifies representation and manipulation of addresses.

Example:

Address: 0001 1010 = Hexadecimal Address: 1A

Advantages and Disadvantages of hexadecimal role in memory addressing :

Advantages:

- It Reduces length of binary numbers by dividing them in 4 bits and makes it easier to read and understand.

Disadvantages:

- This Requires knowledge of the base-16 system, which may be unfamiliar to those use decimal and can make it complex for unfamiliar people.

Octal Number System Preferred Situations in memory addressing :

Octal was once been favored in certain computing environments, particularly in early computer architectures, where a byte (8 bits) could be neatly divided into three octal digits. It can still be useful in specific scenarios such as UNIX file permissions.

Here, 111000101 in binary is equivalent to 705 in octal, where:

$$111 = 7$$

Advantages and Disadvantages

Advantages:

- Provides a more improved representation of binary numbers than binary but is simpler than hexadecimal.
- Example Useful in specific applications like UNIX file permissions (e.g., `chmod 755`).

Disadvantages:

- Less used compared than binary or hexadecimal, leading to reduced familiarity.

- Fewer digits mean that conversions to and from decimal can be less straightforward than with hexadecimal.

Comparison of Number Systems

Summary

In conclusion, binary, hexadecimal, and octal number systems serve distinct purposes in computing. Binary is the foundation for machine-level language, while hexadecimal provides a more human-readable format for representing binary data, especially in memory addressing. Octal has historical significance and specific applications but is less commonly used today. Understanding these number systems is crucial for anyone working in computer science, programming, or related fields. Each system has its advantages and disadvantages, making them more effective.